



BRIEF CASES

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In a Bind: Peak Sealing Technologies' Product Line Extension Dilemma

It was a scorching hot day in July 2013 as Emma Taylor arrived at her office in Peak Sealing Technologies' Dallas, Texas corporate headquarters. Taylor had been with Peak Sealing Technologies (PST) for six years since earning her MBA. She was the product manager for K2-Tape, the company's premium pressure sensitive carton sealing tape line. Recently, a small regional tape producer had found some success with a cheaper, economy grade version that lacked the patented adhesive technology and high quality materials that K2-Tape products offered. Some members of PST's executive steering committee thought the best way to defend against this threat was to add a similar lower quality, less expensive product to the PST line up. PST's president and chief executive officer, Ethan Sloane, was not convinced this was the best strategy and had asked Taylor to provide a formal recommendation on how to deal with the economy product threat.

As Taylor was thinking through the pros and cons of introducing a down-market product, she recalled the speech Sloane had given last month at the Investor Day event:

PST's annual sales growth rate has averaged over 30% over the past five years. This represents unparalleled performance. During the past decade we have led the industry in pressure sensitive tape innovations and advanced adhesive functionality. Our success has been rooted in our technological leadership and intensive R&D efforts. Our future lies in our continued development of cutting-edge technology to provide stronger, more secure, efficient sealing solutions to our customers. PST's mission focuses on providing our customers with the highest quality, most innovative products on the market.

Background and Competitive Overview

In the adhesive industry, PST was recognized as a leader in quality, performance and technological innovation. PST product lines included carton sealing and industrial tapes, specialized adhesives, other packaging products as well as complementary packaging systems and solutions. PST products were generally utilized for packaging and manufacturing applications. Company sales

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and profits had grown significantly over the past 5 years, both organically and through strategic acquisitions. By 2012, net sales were approximately \$240 million with earnings before tax of almost \$35 million. **Exhibit 1** provides historical financial data.

K2-Tape was PST's brand name for its pressure sensitive carton sealing tape line. It was suitable for manual/dispenser applications or automatic machine use. K2-Tape was manufactured using patented adhesive technology with high quality materials and a polyester backing, which resulted in enhanced sealing properties with very high tensile strength¹ and tear resistance². The K2-Tape formulation delivered a higher level of performance versus standard hot melt/BOPP film³ backing that was used by many K2-Tape competitors. K2-Tape's polyester film backing provided more durability, holding force, and higher heat/cold resistance than BOPP film and was suited for more challenging sealing applications. The K2-Tape product value proposition focused on the lowest overall cost to the end user through superior performance. In-house testing provided extensive comparative data illustrating K2-Tape's exceptional performance under wide-ranging conditions.

There were many competitive tape products in the market. Carton sealing tape was available in a variety of strengths, widths, thickness, adhesives and backing materials. K2-Tape offered the most advanced adhesive technology, but often competed against inferior, economy grade products like those produced by Ipack. Ipack tapes were lower quality, BOPP backed products that lacked K2-Tape's strength and consistent performance, but end-user pricing was often 25-35% below K2-Tape. **Exhibit 2** provides end user prices for select Ipack tapes. In 2012, K2-Tape sales were approximately \$68.6 million (**Table A** provides K2-Tape annual sales figures), compared to Ipack sales of \$28.5 million in 2012.

Table A Annual K2-Tape Sales (in Millions)

2008	2009	2010	2011	2012
\$39.6	\$44.4	\$49.9	\$57.5	\$68.6

Tensile, a new Waltham, Massachusetts based competitor, offered an economy BOPP film backed carton sealing tape and had recently entered the Boston metropolitan market with great success. There were also signs of other small regional firms setting up similar economy tape operations in markets throughout the Southeast and Midwest. Taylor had immediately taken notice of this new competition and Tensile's operations in particular. With only one fulltime sales rep, Tensile had reached what equated to a \$2.7 million annual sales rate in the Boston market. A few K2-Tape distributors had already started stocking the Tensile line.

The Adhesive and Sealant Industryⁱ

Adhesive and sealant use was diverse across a wide variety of industries. The industry could be segmented by product type as well as end-user market. Segmentation by product included categories such as natural adhesives, emulsion and dispersion chemistries, hot melt formulations and other reactive and solvent-based adhesives. Although adhesives and sealants were utilized in almost

¹ The force required to break a piece of tape by pulling on opposite ends of the piece.

² The ability of a tape to resist tearing after a tear has been started by cutting or nicking of the edge.

³ Biaxially Oriented Polypropylene (BOPP) is a polypropylene film that has been stretched in two different directions.

every sector of the economy, end-user market demand was often segmented into the following sectors:

- Packaging – Adhesive demand in the packaging industry was at 5,045 million pounds in 2012, a 4% net decline versus 2007. This segment represented a mature market with demand generally driven by the volume of packaging shipments.
- Manufacturing & Assembly – Key industries for the manufacturing and assembly market included motor vehicles, aerospace equipment, electrical and electronic equipment, furniture and machinery. Demand for this segment was at 2,429 million pounds in 2012 with moderate growth expected from a rebound in manufacturing activity.
- Construction – Demand for adhesives in the construction market was 1,053 million pounds in 2012. This segment was forecasted to be the fastest growing sector with 6.1% annual growth projected from 2012 to 2017. Growth was expected to be driven by new building construction and increased repair spending.
- Consumer/Other – This segment comprised a great variety of adhesives for home use such as super glue, wood glue, white glue as well as adhesives used in crafts/hobbies, nail repair, jewelry design and other miscellaneous applications.

US market demand for adhesives typically followed overall economy trends, with major indicators consisting of packaging shipments, manufacturers' shipments and construction expenditures. **Exhibit 3** presents demand estimates and projections for Adhesives and Sealants by market segment and **Exhibit 4** provides forecasts for key indicators driving demand.

K2-Tape

Products: The K2-Tape product line consisted of nine pressure sensitive carton sealing tapes. Products were manufactured with a premium polyester film backing and a patented pressure sensitive hot melt adhesive technology. The majority of K2-Tape end users were in packaging, retail and manufacturing businesses, with specialty customers in military supply and hazardous materials. Applications for K2-Tape products focused on carton sealing, bundling and critical closure applications. Lab tests on technical performance showed exceptional adhesion properties, holding force, tear resistance and tensile strength versus all competitors. **Exhibit 5** provides details and sample applications for the different grades. K2-Tape was known for its reliable seals in high stress shipping situations and superior performance for challenging sealing applications.

In January 2012, K2-Tape introduced GP-48, a general purpose tape. The product was added to offer a more price competitive option for those customers with less demanding needs. GP-48 had the lowest thickness and tensile strength that PST could manufacture with its patented adhesive technology. Case sales by grade for the second half of 2011 and the first half of 2012 are shown in **Table B**.

Table B K2-Tape Case Sales by Grade

Grade	July - December 2011 Case Sales 50m length, 36 roll case	January - June 2012 Case Sales 50m length, 36 roll case
GP - 48	-	12,929
LD - 48	42,943	35,234
RD - 48	55,450	58,838
HD - 48	3,658	3,214
RD - 72	23,177	30,673
HD -72	3,173	2,843
RDS - 72	32,139	31,683
HDS -72	18,304	15,832
SD - 72	2,279	986
Total Sales	181,123	192,232

Pricing: All K2-Tape carton sealing tape products were sold through distributors. The company employed a premium pricing strategy that provided an attractive margin on variable costs. **Table C** provides total variable costs and prices to the distributors.

Table C PST Variable Costs and Distributor Prices (per 50m length, 36 roll case)

Grade	Total Variable Cost	Price to Distributor	PST Dollar Margin
GP - 48	\$ 51.42	\$ 76.87	\$ 25.45
LD - 48	\$ 59.74	\$ 112.87	\$ 53.13
RD - 48	\$ 76.72	\$ 162.31	\$ 85.59
HD - 48	\$ 121.16	\$ 210.08	\$ 88.92
RD - 72	\$ 114.37	\$ 191.79	\$ 77.42
HD -72	\$ 142.24	\$ 243.85	\$ 101.61
RDS - 72	\$ 135.49	\$ 243.85	\$ 108.36
HDS -72	\$ 165.86	\$ 293.29	\$ 127.43
SD - 72	\$ 264.22	\$ 525.75	\$ 261.53

The company sales force provided guidance on end-user pricing through suggested resale prices provided in **Exhibit 6**. PST had a very selective distribution policy and therefore distributors generally complied with the suggested prices.

Sales Force: PST maintained a 40-person corporate sales force that was responsible for selling K2-Tape as well as other PST products.

The abundance and variety of carton sealing tape products often caused confusion among end users with no standard criterion for procurement decisions. Some customers put a great deal of thought into purchase decisions, considering tape performance properties compared to needs, while others purchased the same tape regardless of the application. Based on her research, Taylor estimated that packaging engineers drove approximately 40% of tape procurement decisions. These buyers best understood the value of premium products and the cost savings that could come from using K2-Tape. They utilized technical performance testing and product specifications to find the

right tape for each application. However, Taylor also knew there were many customers at the other end of the spectrum whose first and foremost purchase consideration was simply price per roll, with the next consideration being delivery dependability.

PST's sales force spent the majority of their time working with end users, finding solutions to complex adhesion problems, analyzing applications on-site and performing cost studies to demonstrate the cost effective nature of PST products. PST corporate research labs supported this function by demonstrating how different products helped decrease material and labor costs and reduce damage related to adhesive failure. With carton sealing tape, the sales force often had to sell against economy grade products. As a result, the company had developed specific training materials and marketing collateral to address the advantages of K2-Tape's patented polyester backed adhesive technology and how to position products against economy grade competitors. For example, economy tape could fail as a result of excessive heat or cold which could result in tape brittleness, shrinkage, adhesive decomposition and other degradation of sealing properties. In addition, economy tape used on packages that were too heavy, had unstable contents, long shipment cycles or multiple handlings were also at high risk of failure. Tape failure could be extremely costly, resulting in damaged or lost contents as well as buyer dissatisfaction. K2-Tape users could also often use less product compared to economy, while obtaining a more secure seal. K2-Tape had achieved solid sales growth in the recent economic downturn by seeking out new customers and educating them on how high performance products could lower overall expenses.

Sloane often cited the analytical sales approach as a key factor in the company's success. He stated, "Our sellers are the engine of our growth strategy. When we truly understand our clients' challenges and can demonstrate the benefits of superior performance, we deliver the most compelling value proposition. There is no question that PST has the best trained and most knowledgeable sales force in the industry."

The acquisition of a packaging systems firm in 2009 had diverted some of the sales force attention from K2-Tape in recent years. The packaging systems and solutions division's main product lines included bagging and shrink wrap systems, tape dispensers, and automated carton forming/closure equipment systems that were compatible with K2-Tape products. Gross margins for packaging systems were normally higher than for K2-Tape products. Prior to the acquisition, a 22 person sales force devoted 85% of their time to K2-Tape products. Following the acquisition, a sales force of 40 split their time among all products. Approximately 35% of their time was allocated to K2-Tape, 60% of time was spent on packaging systems and solutions and 5% was allocated to the other PST products.

Sales force compensation was above industry average. Compensation structure was comprised of a base salary plus commission of 2% for K2-Tape sales and 1% net sales for all other products with attractive incentive bonuses for packaging systems sold.

Distributors: PST had very strong relationships with its highly selective distributor network. In 2012, the corporate distribution network included 125 distributors with 55 of these designated as "Distinctive Distributors" because together they handled over 80% of the business. The largest 10 K2-Tape distributors represented approximately 35% of sales.

Distributors usually stocked a wide range of products with many large distributors providing one stop shopping for all packaging needs. Most distributors carried competitor products, but had a history of steering customers to PST products. In return for this distributor loyalty, PST maintained its limited, selective distribution policy. Some distributors had previously requested an PST economy tape line, but others complained there were already too many grades.

Distributors' main function for K2-Tape was order processing and fulfillment. Distributors typically did not demonstrate features to customers. When distributors identified potential new K2-Tape business, they would refer that customer to the PST sales team and arrange a joint call. Some distributors had recently complained that PST sales teams were too busy with other products and did not spend enough time servicing existing K2-Tape customers and developing new business. Distributor satisfaction surveys for 2012 indicated 20% lower satisfaction scores for K2-Tape sales force efforts as compared to 2009.

Competitive Threat

Tensile's BOPP economy product line featured two items and had quickly gained a foothold in the Boston metropolitan market. Taylor had not been surprised that the Boston market was so open to a new low-price product. It was acknowledged in the industry that many Boston customers were extremely price sensitive. PST's largest distributor in the Northeast commented "The Boston market is known for its lack of brand loyalty. Some customers are willing to switch brands and distributors based on price variances of only a few dollars on a large order."

Tensile distributor prices and suggested resale prices to end users for the Boston market are shown in **Table D**:

Table D Tensile Distributor Prices (per 50m length, 36 roll case)

	<u>Ecoslim</u>	<u>Ecoplus Wide</u>
Price to Distributor	\$ 118.02	\$ 134.44
Suggested Resale	\$ 159.85	\$ 212.95

PST Northeast regional sales manager, Anthony Robinson, estimated Tensile annual sales would be about \$2 million for Ecoplus Wide (72mm width with 2.4 mils thickness) and approximately \$675,000 for Ecoslim (48mm width with 2.2 mils thickness).

Taylor had researched the product specs and lab results for the more established economy competitor, Ipack, and was just starting to gather data on Tensile. While Taylor always considered Ipack's products substandard with test results proving Ipack's vast inferiority, she had to admit that a preliminary look at Tensile showed they were actually halfway decent for uncomplicated, regular-duty applications. More extensive tests would need to be performed, but her initial opinion was that for the simplest applications, Tensile's Ecoslim might become a competitor to K2-Tape's RD-48 and Ecoplus wide could challenge RDS-72. Although Taylor considered the Tensile products respectable for BOPP economy grade, she expected their performance would deteriorate significantly for anything beyond the most undemanding regular-duty application. Robinson believed Tensile's sales success to date came at the expense of Ipack, not K2-Tape.

Maintain Premium Product Line or Follow Down-Market?

PST had explored entering the economy grade space previously. Given the competitive situation at the time, PST decided not to enter. But now Tensile prompted reconsideration of the question. A new PST economy product could utilize a simpler, non-patented adhesive formulation with BOPP backing. Extensive research had been done on production requirements and cost estimates. The head of manufacturing had confirmed there was suitable equipment and capacity at the Atlanta

plant, and production could ramp up quickly. There would be minimal start-up expenses with no major capital investment required. The preliminary estimate for total variable costs per 50m, 36-roll case was \$80.75 for a competitor to Ecoslim and \$107.95 for competitor to Ecoplus Wide. The company believed these estimates were similar to Tensile's variable costs.

Taylor had to decide whether to recommend the company stay the course with advanced technology, high-quality products or enter the economy grade segment. During Ethan Sloane's tenure, PST had successfully defended against a diverse set of competitors. However, Taylor knew the packaging tape industry was a mature market and competition was heating up. Was this a threat they should respond to? Could a company known as a technology leader in premium grades offer an economy product? If they were to introduce an economy grade, how should it be marketed? Taylor settled into her chair and started to outline her options and the implications for their customers, distributors, sales force and brand. Her report for Sloane was due by the end of the week.

Exhibit 1 Selected PST Financial Data (\$ in thousands)

	2008	2009	2010	2011	2012
Net Sales by class of product					
Adhesive Tapes (Carton Sealing and Industrial)	\$ 51,143	\$ 58,054	\$ 67,826	\$ 81,289	\$ 93,034
Packaging Systems/Solutions	\$ 8,263	\$ 41,975	\$ 57,270	\$ 78,742	\$ 105,153
Other Adhesives	\$ 12,339	\$ 9,742	\$ 9,358	\$ 9,301	\$ 9,994
Other Packaging	\$ -	\$ 7,268	\$ 12,585	\$ 21,547	\$ 31,916
Total Sales	\$ 71,745	\$ 117,039	\$ 147,039	\$ 190,879	\$ 240,097
Costs and expenses					
Cost of sales	\$ 44,582	\$ 65,772	\$ 84,311	\$ 117,069	\$ 146,679
Marketing, admin, development	\$ 18,146	\$ 32,772	\$ 39,368	\$ 45,677	\$ 58,224
Other income (expenses)	\$ 87	\$ (2,211)	\$ (2,000)	\$ (753)	\$ (322)
Earnings before income tax	\$ 9,104	\$ 16,284	\$ 21,360	\$ 27,380	\$ 34,872

Exhibit 2 Suggested End-User Prices (per 50m length, 36 roll Case) for Select Ipack Products

Economy Carton Sealing Tape (1.5 Mil)	\$ 72.88
Economy Plus Carton Sealing Tape (2.2 mil)	\$ 160.45
Industrial Light Duty Economy Tape (2 mil)	\$ 109.47
Heavy Duty Economy Industrial Tape (2.4 mil)	\$ 230.84

Exhibit 3 United States: Adhesives & Sealants Demand by Market (million pounds)^a

	2007	2012	Projected 2017	CAGR % 12/07	17/12
Adhesive & Sealants Demand	9,355	8,750	9,775	-1.3%	2.2%
Packaging	5,250	5,045	5,330	-0.8%	1.1%
Manufacturing & Assembly	2,505	2,429	2,795	-0.6%	2.8%
Construction	1,362	1,053	1,415	-5.0%	6.1%
Consumer/Other	238	223	235	-1.3%	1.1%

^aThe Freedonia Group, Inc., *Adhesive & Sealants: United States* March 2013; pg. 15

Exhibit 4 United States: Key Indicators for Adhesives & Sealants Demand (billion 2005 dollars)^a

	2007	2012	Projected 2017	CAGR % 12/07	17/12
Gross Domestic Product	13,206	13,600	15,500	0.6%	2.6%
Resident Population (million persons)	301.2	314.0	326.3	0.8%	0.8%
Packaging Shipments	116.0	105.5	113.5	-1.9%	1.5%
Manufacturers' Shipments	4,617	4,210	4,640	-1.8%	2.0%
Construction Expenditures	1,069	725	1,060	-7.5%	7.9%

^aThe Freedonia Group, Inc., *Adhesive & Sealants: United States* March 2013; pg. 12

Exhibit 5 Sample Applications by Grade for K2-Tape Products

Product	Grade	Thickness	Width	Application Example
GP - 48	General Purpose	1.7 mil	48mm	Packages less than 15 pounds with stable contents and inexpensive value
LD - 48	Light Duty	2.0 mil	48mm	Packages less than 20 pounds with minimal shifting of contents and inexpensive value
RD - 48	Regular Duty	2.2 mil	48mm	Packages 15-30 pounds with minimal shifting of contents and moderate value
HD - 48	Heavy Duty	2.5 mil	48mm	Packages over 30 pounds with minimal shifting of contents and high value
RD - 72	Regular Duty	2.2 mil	72mm	Packages 15-30 pounds, size of over 30" x 22" with minimal shifting of contents and moderate value
HD - 72	Heavy Duty	2.5 mil	72mm	Packages over 30 pounds, size of over 30" x 22" with minimal shifting of contents and high value
RDS - 72	Regular Duty Plus	2.5 mil	72mm	Packages 15-30 pounds, size of over 30" x 22" with unstable contents shifting and moderate value
HDS - 72	Heavy Duty Plus	3.0 mil	72mm	Packages over 30 pounds, size of over 30" x 22" with unstable contents and high value
SD - 72	Super Heavy Duty Plus/Security	3.4 mil	72mm	Packages over 30 pounds, size of over 30" x 22" with unstable contents and high value needing security tape features

Exhibit 6 K2-Tape Suggested Resale Price List (per 50m length, 36 roll Case), Effective March 2012

Grade	Price	
GP - 48	\$	95.90
LD - 48	\$	142.17
RD - 48	\$	200.56
HD - 48	\$	261.94
RD - 72	\$	239.74
HD - 72	\$	303.73
RDS - 72	\$	303.73
HDS - 72	\$	367.73
SD - 72	\$	655.04

Endnotes

ⁱ Freedonia Group, Inc., *Adhesive & Sealants: United States* March 2013; pgs. 2-16